

RAMBØLL	PROJECT —	PROJECT NO. —	PREPARED —
	Simply Supported Steel Beam — quick check		CHECKED —
REV / DATE 0.1 · 2026-07-20			APPROVED —
			DRAFT

1 Input Parameters

Symbol	Value	Unit	Description
$beam_{h1}$	1		beam_h1
$Span_L$	6	m	Span L (m)
$Line_{load, w}$	12	kN/m	Line load w (kN/m)
$Section_{modulus, Wel}$	324	cm3	Section modulus Wel (cm3)
$Yield_{fy}$	355	MPa	Yield fy (MPa)
$factors_{b2}$	1		factors_b2

2 Calculation

Moment M (kNm)

$$Moment_M = \frac{Line_{load, w} \cdot Span_L^2}{8}$$
$$= \frac{12 \cdot 6^2}{8}$$

= 54 kNm

Bending stress sigma (MPa)

$$Bending_{stress, \sigma} = Moment_M \cdot 1000 \cdot \frac{1}{Section_{modulus, Wel}} \cdot factors_{b2}$$
$$= 54 \cdot 1000 \cdot \frac{1}{324} \cdot 1$$

= 166.7 MPa

3 Verification

OK

$$Bending_{stress, \sigma} = 166.7 \text{ MPa} \leq Yield_{fy} = 355 \text{ MPa}$$